Statistics

Introduction — Population, Sample & Variables

28 March 2026 · Statistics Introduction

What is Statistics | Key Terms | Population vs Sample | Variables

§ 01 What is Statistics?

DEFINITION

Statistics is a domain of Mathematics / Science where we focus on **collecting, organising, analysing & interpreting data** to make decisions.

Real-world Questions Statistics Answers:

→ What is the average height in the class?

→ What is the average age in the class?

→ What is the chance it will rain tomorrow?

→ If I do more study, will I get better grades?

§ 02 Key Terms

Term	Definition	Example
Population	The entire group we want to study	All 1000 students in Batch-2
Sample	A small part of the population we actually observe	100 live students

Parameter	A number that describes the population	Mean age of all 1000 students
Statistic	A number that describes the sample	Mean age of 100 sampled students
Variable	Something we measure	Height, weight, marks
Observation	One single data point (one row)	Student s: height=h
Data	Collection of all observations	Table with name, height, weight, age

§ 03 Population vs Sample

CORE IDEA

Population = the entire group (all students, all customers, all data points).

Sample = a smaller subset selected from the population.

Bigger the sample → **better the estimate** of the population.

Describing population using its sample → use **statistics**.

Describing population using its full data → use **parameters**.

Batch-2 Example:

	Population	Sample
Group	1000 all students	100 students (live class)
Avg Age	Describes the population	Describes the population from sample
Fresher%	Describes the population	Describes population from sample
Called	**Parameter**	**Statistic**

RULE

"Bigger the sample, better the estimate."

Example: 2 marks → avg=10. 100 marks → avg = 10,10,20,30... → better estimate.

Like a bag of marbles — handful vs full bag: handful gives rough idea, bigger sample = better.

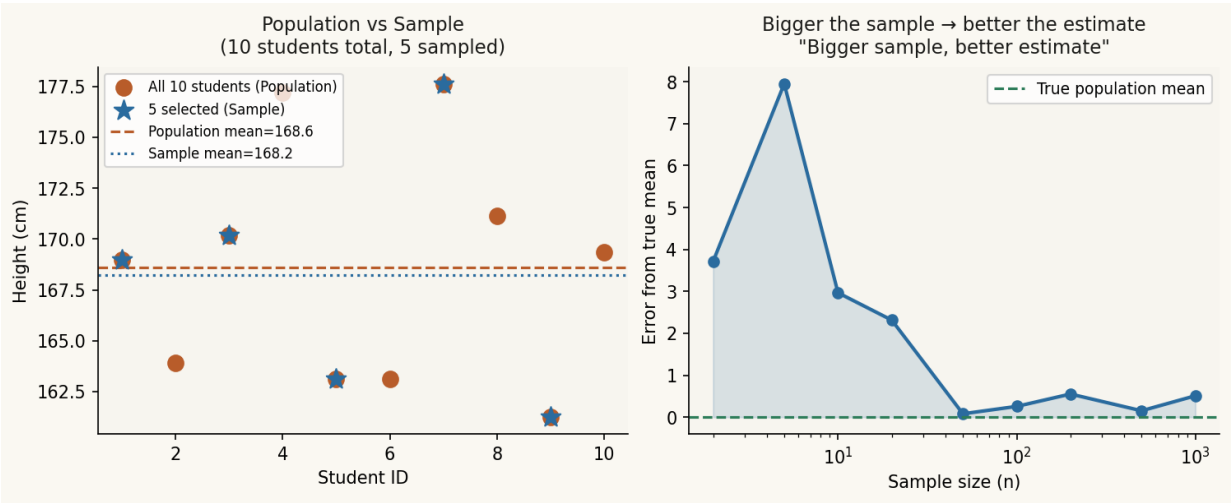

Figure 1: Population vs Sample — scatter plot and sample-size effect on estimate accuracy

§ 04 Types of Variables

TWO TYPES

A **variable** is something we measure or observe. There are 2 types:

1. **Qualitative (Categorical)** — describes a category, not a number
2. **Quantitative (Numerical)** — a measurable number (how much?)

Type	Also called	Key Question	Examples
Qualitative	Categorical	What kind / which category?	Gender, city, class, subject, email

Quantitative	Numerical	How much / how many?	Age, height, marks, temperature, battery %, salary, distance

T-shirt Brand Example (Practical Application):

A clothing brand sells T-shirts: XS, S, M, L, XL, XXL

They collect **height (h)** and **weight (w)** of customers → these are **quantitative variables**.

They use a **sample** of customers to decide how many of each size to produce → sample statistics describe the population.

Figure 2: T-shirt brand example — sample mirrors population proportions across sizes

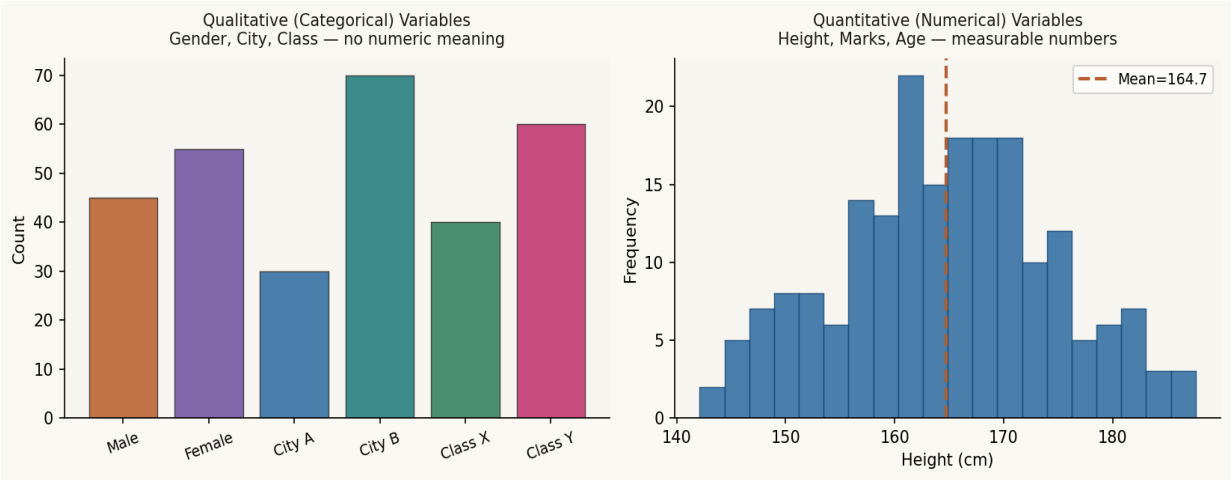

Figure 3: Qualitative (categorical bar chart) vs Quantitative (numerical histogram)

§ 05 Summary

Concept	Key Point
Statistics	Collect, organise, analyse, interpret data → make decisions
Population	Entire group | described by Parameters
Sample	Subset of population | described by Statistics
Bigger sample	→ Better estimate of population parameter
Variable	Something we measure — 2 types: Qualitative & Quantitative
Qualitative	Category (Gender, City, Class) — no numeric meaning
Quantitative	Number (Age, Height, Marks) — how much / how many
Observation	One data point = one row in a table
Data	Collection of all observations (name, h, w, age…)

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